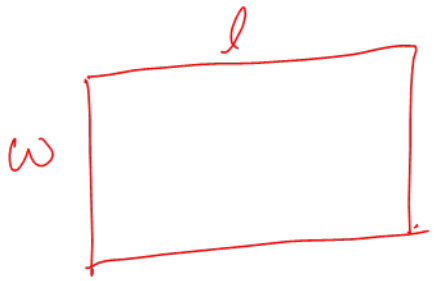
Goal: Use the tools used to find maximums and minimums in applications.

Main Idea: Given a situation, find the "optimal" value (usually a maximum or minimum).

Example 1. A farmer is building a rectangular fence and has 400′ of fencing to work with. What should the dimensions of the fence be to make the area as large as possible?

400ft

The function we want to optimize is called the objective function. What is it for this example?



Maximize $A_{(\ell,w)} = \ell w$

In order to reduce the number of variables down to 1 and/or guarantee a max/min, we need more conditions called Constraints. Are there any in this example? "ha[ve] 400' of fencing"

given $2\ell + 2w = 400$

$2\ell = 400 - 2w$

$\ell = 200 - w$

We can now finish the problem (by finding the absolute max).

$A(\ell, w) = \ell w$

$A(w) = (200 - w)w$

Find all possible locations of an abs. max

$A(w) = 200w - w^2$

$$\frac{dA}{dw} = 200 - 2w \overset{\text{set}}{=} 0$$

$$200 = 2w$$

$$w = 100$$

Verify A has an abs. max at $w = 100$.

$$\frac{d^2A}{dw^2} = -2 < 0$$

Rel. Max by 2nd $\frac{d}{dx}$ test

+

Only One Critical number $\Longrightarrow$ Abs. Max at $w = 100$

Answer the question:

Ie, we also need to find the length

$$l = 200 - w$$

$$l\big|_{w=100} = 200 - 100 = 100$$

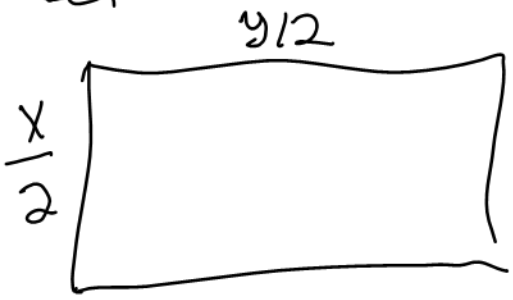
Therefore, the area of the enclosure is maximized when it is a square with lengths of 100'.

$$x + y = 5$$

Maximize the product xy

Example 2. Suppose that two numbers x and y add up to $S > 0$. When is their product at a maximum?

Ie, we want to "maximize area"



Obj: Max $A(x,y) = xy$

Given: $x + y = S$

$y = S - x$

$$A(x,y) = xy$$
$$A(x) = x(S-x)$$
$$A(x) = Sx - x^2$$
$$\frac{dA}{dx} = S - 2x \quad \text{set } 0$$
$$S = 2x$$
$$x = \frac{S}{2}$$

Verify it is the loc. of the abs. max.

$$\frac{d^2A}{dx^2} = -2 < 0$$

Ie, since there is only 1 crit. num, there is an abs. max at $x = \frac{S}{2}$ by the 2nd $\frac{d}{dx}$ test.

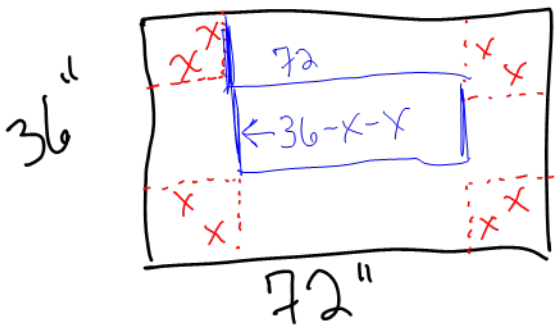
Find y: $y = S - x$

$$y\Big|_{\frac{S}{2}} = S - \frac{S}{2} = \frac{S}{2}$$

Conclusion: The product is maximized when

$$X = Y = \frac{S}{2}$$

Example 3. A $36'' \times 72''$ piece of cardboard is cut from each corner and the sides are folded up to create a box with an open top. How far from the edge should the cut be to maximize the volume?



Obj: Max $V(x) = (36-2x)(72-2x)x$

Given: $0 \le x \le 18 \Leftarrow x \in [0,18]$

$$V(x) = (36-2x)(72-2x)x$$

$$V(x) = 2592x - 216x^2 + 4x^3$$

$$\frac{dV}{dx} = 2592 - 432x + 12x^2 \overset{\text{set}}{=} 0$$

$$ax^2+bx+c=0 \Rightarrow x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$x = 18 \pm 6\sqrt{3}$$

$x \in [0,18] \Rightarrow x = 18 - 6\sqrt{3} \approx 7.61$

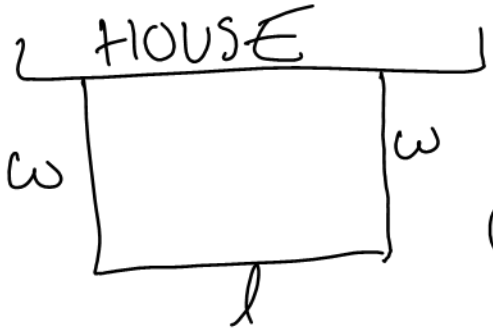
Find the max of V on $[0,18]$

x	0		$18-6\sqrt{3}$		18
$(36-2x)(72-2x)x$	0		>0		0

Ie, the max occurs at $x = 18-6\sqrt{3}$

Therefore, we only need to cut roughly 7.61" from the edge to maximize volume.

Example 4. You are making a rectangular garden and you have 50′ of fencing to work with. To make a bigger garden, you build it right next to your house. How large can the garden be?



Obj: Max $A(l, w) = lw$

Given: $2w + l = 50$

$l = 50 - 2w$

$A(l, w) = lw$

$\Rightarrow A(w) = (50 - 2w)w$

$A(w) = 50w - 2w^2$

$\frac{dA}{dw} = 50 - 4w \overset{\text{set}}{=} 0$

$w = \frac{50}{4} = \frac{25}{2}$

Verify it's a max:

$\left.\frac{d^2A}{dw^2}\right|_{w=\frac{25}{2}} = -4 < 0$

$\Rightarrow$ Abs. max at $w = \frac{25}{2}$ by the 2nd $\frac{d}{dx}$ test

To answer the question, we need to compute

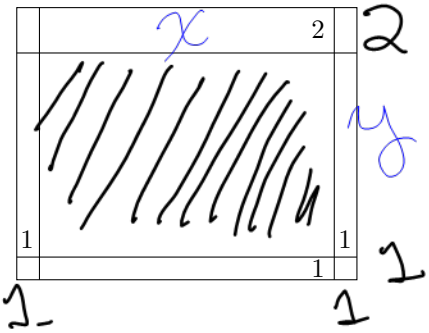
$A\left(\frac{25}{2}\right) = \left(50 - 2\left(\frac{25}{2}\right)\right)\left(\frac{25}{2}\right)$

$= (50 - 25)\left(\frac{25}{2}\right) = \frac{25^2}{2}$

$= 312.5 \text{ ft}^2$

The maximum area is 312.5 ft^2

Example 5. A poster board has an area of $180\,\text{cm}^2$ and has margins as shown in the diagram below. What is the largest printable area?



Obj: Maximize

$A = xy$

Given:

$(x+1+1)(y+2+1) = 180$ AND $x, y \geq 0$

$(x+2)(y+3) = 180$

$x+2 = \dfrac{180}{y+3}$

$x = \dfrac{180}{y+3} - 2$

$A(x,y) = xy$

$\rightsquigarrow A(y) = \left(\dfrac{180}{y+3} - 2\right)y$

$A(y) = \dfrac{180y}{y+3} - 2y$

$\dfrac{dA}{dy} = \dfrac{180(y+3) - 180y}{(y+3)^2} - 2$

$= \dfrac{180y + 540 - 180y}{(y+3)^2} - 2$

$$\frac{dA}{dy} = \frac{540}{(y+3)^2} - 2 \overset{\text{set}}{=} 0$$

$$\frac{540}{(y+3)^2} = 2$$

$$540 = 2(y+3)^2$$

$$270 = (y+3)^2$$

$$y + 3 = \pm\sqrt{270}$$ ← 9:30

$y \geq 0$ →

$$y = -3 \pm 3\sqrt{30}$$

$$y = -3 + 3\sqrt{30}$$

$$A = (-3 + 3\sqrt{30})(2\sqrt{30} - 2)$$

$$\approx 120.3 \text{ cm}^2$$